DTNet: A Specialized Dual-Tuning Network for Infrared Vehicle Detection in Aerial Images

Nan Zhang, Youmeng Liu, Hao Liu, Tian Tian, Jiayi Ma, *Senior Member, IEEE*, and Jinwen Tian

***Abstract*— Vehicle detection in infrared aerial images is vital for both military and civilian applications, as infrared imaging remains effective under low-light conditions and various adverse weather scenarios. However, the longer wavelengths of long-wave infrared, compared to visible light, make diffraction more noticeable, leading to low-frequency degradation of vehicle information. Thermal radiation from the environment and optical system leads to higher noise in infrared images. Additionally, atmospheric transport models for various weather conditions can degrade infrared images to different extents. These factors lead to a reduced signal-to-noise ratio (SNR), which complicates the extraction of clear features. To overcome these challenges, we propose the dual-tuning network (DTNet), an advanced framework for vehicle detection in infrared aerial images, developed based on the mechanisms of infrared imaging. Specifically, the core component of DTNet is the dual-tuning block (DTBlock), which works alongside the dynamic guided filtering module (DGFM) and the point spread recovery module (PSRM) for feature extraction. DTBlock decomposes feature maps into low- and high-frequency components with learnable low-pass filters. DGFM eliminates disturbance from the optical system and background thermal radiation in the high-frequency component of feature maps, while preserving the details and texture of vehicles. PSRM aggregates vehicle features in the low-frequency component of feature maps, which is proposed with reference to diffraction and atmospheric models. The concept of dual-tuning refers to enhancing the signal and suppressing interference in the low- and high-frequency parts, respectively. Experimental results on the DroneVehicle public dataset for infrared vehicle detection indicate that our proposed approach achieves state-of-the-art (SOTA) performance. Moreover, extensive ablation studies confirm the superior capability of our DTNet in robust feature extraction from infrared images.**

***Index Terms*— Feature extraction, frequency decomposition, image degradation, infrared aerial image, vehicle detection.**

I. Introduction

INFRARED vehicle detection is a significant and challenging field in remote sensing. This task holds immense importance across numerous applications, including military target identification [1], analysis of traffic dynamics [2], [3], [4], [5], [6], emergency rescue operations, and security measures [7], [8], which has made it a focal point of interest among the research community. Long-wave infrared offers advantages over visible light, such as the ability to image in low-light conditions and to penetrate slight haze. However, it experiences more severe low-frequency degradation, which manifests as blurred details and reduced contrast [9]. This degradation is primarily due to energy diffusion during transmission through the atmosphere and apertures [10]. Additionally, environmental stray radiation and various optical system noises, such as thermal noise from electronic components and shot noise from amplification circuits, increase the noise level in infrared images [11]. While infrared and visible light images share some common causes of degradation, their degradation models differ. In the atmosphere, long-wave infrared primarily attenuates due to absorption and re-radiation, whereas visible light mainly attenuates due to scattering [12], [13]. The longer wavelengths of long-wave infrared also result in more significant diffraction when passing through apertures [10]. Moreover, single-band long-wave infrared contains less information, thus requiring more attention to low-frequency degradation and interference in infrared images.

Nearly all existing detection approaches proposed for infrared images concentrate on their superficial characteristics, such as blurred edges, low contrast, and clutter noise. Most of them enhance the network by incorporating additional information, including implementing pixel-level supervision to refine object contours [9], [14], integrating multimodal inputs like visible images [15], [16] and point clouds [5] to enrich detail, and employing inter-frame continuity to mitigate noise interference [17], [18], [19], [20]. However, these efforts often do not significantly improve the ability of networks to extract features from infrared images. Because they address only the symptoms rather than the underlying issues, lacking analysis of the factors contributing to degradation in infrared images.

In this article, we attempt to model the long-wave infrared imaging process to analyze the factors of degradation, which informed the development of dual-tuning network (DTNet), a novel network family tailor-made for vehicle detection from infrared images. DTNet integrates a core dual-tuning block (DTBlock) with two modules: the dynamic guided filtering module (DGFM) and the point spread recovery module (PSRM). We will further elaborate on the design and

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Nan Zhang, Youmeng Liu, Hao Liu, Tian Tian, and Jinwen Tian are with the National Key Laboratory of Multispectral Information Intelligent Processing Technology, School of Artificial Intelligence and Automation, Huazhong University of Science and Technology, Wuhan 430074, China (e-mail: zhang_nan@hust.edu.cn; ym_liu@hust.edu.cn; hao_liu@hust.edu.cn; ttian@hust.edu.cn; jwtian@mail.hust.edu.cn).

Jiayi Ma is with the Electronic Information School, Wuhan University, Wuhan 430072, China (e-mail: jyma2010@gmail.com).

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rationale of these components. Our analysis identifies atmospheric effects, diffraction, and various types of noise as key contributors to infrared image degradation. We conceptualize atmospheric effects and diffraction as point spread functions that disperse long-wave infrared energy. The superposition of various noises is regarded as a broadband noise. These aspects are detailed in Section III-A. Therefore, DTNet is designed to remove broadband noise and recover radiant energy dispersed. Broadband noise with its randomness and wide spectrum is challenging to eliminate directly. It tends to causes more distortion in the high-frequency component of infrared images, where details are scarce and precious [9]. Point spread functions decrease the frequency of discriminative information about vehicles in infrared images, as observed in the study [9]. The DTBlock within DTNet addresses these challenges by decomposing feature maps into high- and low-frequency components. For the high-frequency component, DGFM is employed to effectively remove high-frequency noise while preserving crucial details in images. The PSRM is applied to optimize the low-frequency components of feature maps, which alleviates the low-frequency degradation and the low-frequency interference. A key element of the PSRM is the integration of the adaptive pyramidal convolution (APConv), which is a specialized feature extraction operator devised from atmospheric effects and diffraction models. APConv uses pyramid receptive fields for point spread areas and the adaptive kernel generation for various atmospheric conditions. In addition, APConv incorporates a global-guided strategy that, when combined with frequency decomposition of the DTBlock, facilitates precise aggregation of energy and efficient interference elimination in the frequency domain.

The contributions of this article are summarized as follows.

1) We explore the factors that lead to the degradation of infrared images during the imaging process, laying the groundwork for designing a network tailored for vehicle detection in infrared aerial images.
2) The DGFM is developed to eliminate high-frequency noise from infrared images while maintaining edges of vehicles.
3) We devise the PSRM to recover the diffusion of thermal radiation caused by diffraction and atmospheric effects, guided by oriented bounding box supervision. The PSRM effectively aggregates crucial low-frequency features and filters out the low-frequency components of broadband noise.
4) The DTBlock is proposed as a fundamental component of our network, which decomposes feature maps into high- and low-frequency parts, and refines these two parts with DGFM and PSRM, respectively.

The remainder of this article is structured as follows. Section II reviews related works on vehicle detection, infrared object detection, and state-of-the-art (SOTA) feature extraction networks. The infrared imaging model and detailed DTNet architecture are presented in Section III. Experimental results and discussions are provided in Section IV, followed by conclusions in Section V.

II. Related Work

A. Vehicle Detection

The prosperity of generic object detection prompts many studies [21], [22], [23], [24] to apply them to vehicle detection. In order to accurately detect densely arranged and arbitrarily oriented vehicles and rectangular targets, oriented bounding boxes are introduced into detectors. The improvement of oriented object detection is primarily divided into two aspects: angle optimization and feature alignment. Early methods adopt a horizontal rectangular box with an angle [25], facing challenges with angular periodicity and boundary loss changes. Addressing these issues, the angular prediction is shifted from regression to classification [26] or an IoU-related factor is introduced in the loss [27], [28], [29]. Yu and Da [30] propose a novel approach by mapping rotational periodicity into different frequency phases. To reduce the complexity of angle optimization, additional exploration in representation includes Gaussian distribution models [31], [32], [33], [34], point set methods [35], [36], and polar coordinates [37], [38]. Some two-stage approaches [29], [39], [40] modify the RPN to generate rotated proposal boxes and use ROI Pooling to align the rotated features. Han et al. [41] employ deformable convolution [42] to effectively capture features of oriented objects. Hou et al. [43] implement diverse sample selection strategies based on aspect ratios to keep the accuracy of the extracted features. However, these approaches, initially designed for visible images, fall short in infrared image applications due to inherent characteristics such as blurred details, low contrast, and broadband noise. Therefore, it is essential to design models suitable for vehicle detection in infrared images by analyzing the properties of infrared imaging.

B. Infrared Object Detection

The challenge of object detection in infrared images is the scarcity of discriminable information, which is reflected in blurred detail and low contrast. Chen et al. [44] enhance the contrast by using morphology cap transformation to sharpen vehicle outlines. Du et al. [45] tackle the challenge of insufficient infrared data by applying transfer learning from visible to infrared datasets and mining negative samples. For infrared dim-small object detection, Zhang et al. [14] introduce edge supervision to enable the network to predict sharp edges. Motion features of the target across multiple frames are introduced to filter out false alarms caused by noise in infrared images [17], [18], [19], [20]. Zhang et al. [9] supplement the detection network with additional semantic information. Moreover, multimodal data fusion is also commonly used to enrich discriminable features. For instance, Sun et al. [15] and Dhanaraj et al. [16] employ fusion techniques that combine visible and infrared images to improve the accuracy of vehicle detection. Yang et al. [5] integrate LiDAR point cloud clusters into infrared object detection to provide additional information about object position and size. These approaches require extra data or annotations to supplement information for infrared images, yet they do not improve the power of networks to extract features, merely addressing the symptoms. On the contrary, we thoroughly analyze the degradation factors in

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infrared images and develop proprietary blocks to eliminate the interference results.

C. SOTA Networks for Feature Extraction

Since the development of deep neural networks, various network architectures and blocks have been proposed to improve the ability of feature extraction. Typical convolutional neural network architectures include VGG [46], ResNet [47], Inception [48], etc. Based on these, numerous powerful networks have been developed for generic vision tasks, such as SKNet [49], ConvNeXt [50], and InceptionNeXt [51]. Benefiting from the Transformer in NLP, ViT [52] and Swin Transformer [53] become powerful model families for vision tasks. In order to explore the key factors contributing to network feature extraction, some works [49], [50], [54], [55] discuss the benefit of long-distance connections. PoolFormer [56] demonstrates that the architecture of the transformer is key to improved performance. Other studies [57], [58], [59] examine the improvement in feature extraction capability due to dynamic weighting. Subsequently, there are many networks that integrate the advantages of convolution and transformer, such as EMO [60] and Next-Vit [61]. Researches on networks not only focus on generic vision tasks but also extend to various application requirements. For strong correlation between target and background in remote sensing images, LSKNet [54] is proposed for modeling contextual associations. The kaleidoscope module, a hybrid of convolution and attention, is proposed in the I^2MDet [9] for extracting features in infrared images. UniRepLKNet [55] establishes a unified CNN architecture for modeling both multimodal and temporal sequential information. MNet [62] integrates 2-D and 3-D knowledge for medical image segmentation. Various refined networks, such as MobileNet [63], FasterNet [64], and EfficientVit [65], are proposed to satisfy the demand for the deployment. However, there is no authoritative network for vehicle detection tasks in infrared images. Therefore, we take into account the challenges inherent in this task and devise a specialized network family for vehicle detection in infrared images.

III. Methods

A. Infrared Imaging Model

The transformation of long-wave infrared emitted by objects in the field of view goes through two processes: propagation in the atmospheric medium and imaging in the optical system. The long-wave infrared is attenuated by the absorption, reradiation and scattering of various particles as it propagates through the air medium. The stray radiation from the environment outside the field of view enters the optical system through reflection and scattering. The radiation entering the optical system is presented as a digital image through the processes of pinhole imaging, sensor photoelectric conversion, electrical signal amplification and sampling. During these processes, diffraction significantly disperses the radiant energy when long-wave infrared passes through the aperture, which becomes more noticeable with wavelength. In addition, various noises resulting from the optical system are also presented on the digital image [10], [66], [67], [68]. Based on the above, the imaging model is shown in (1)

$$\begin{aligned} R_{\text{show}} &= O(\lambda, r)[R_{\text{real}}(\lambda, t)\tau(\lambda, d, w) + R_{\text{env}}(\lambda, t)] + N(\lambda, t) \\ &= R_{\text{real}}(\lambda, t) \cdot O(\lambda, r) \cdot \tau(\lambda, d, w) \\ &\quad + R_{\text{env}}(\lambda, t) \cdot O(\lambda, r) + N(\lambda, t) \\ &= R_{\text{real}}(\lambda, t) \cdot O(\lambda, r) \cdot \tau(\lambda, d, w) + M(\lambda, r, t). \end{aligned} \tag{1}$$

In this model, R_{show} denotes the digital image presented by the device. $R_{\text{real}}(\lambda, t)$ represents the radiation from objects in the field of view, where λ is wavelength and t stands for temperature, in accordance with Planck's law [69]. $R_{\text{env}}(\lambda, t)$ refers to the stray radiation that enters the field of view from the surroundings, which is also characterized by λ and t. The term $\tau(\lambda, d, w)$ defines the point diffusion function of the atmospheric effects, where d denotes the imaging distance and w means the weather conditions. $O(\lambda, r)$ denotes the point spread function of diffraction in the optical system, with r being the radius of the aperture, based on Fraunhofer diffraction theory [70]. $N(\lambda, t)$ signifies the superposition of various additive noises in the optical system, which include thermal noise from electronics, scattering noise from amplification circuits, and nonuniform noise in the focal plane. Since both $R_{\text{env}}(\lambda, t) \cdot O(\lambda, r)$ and $N(\lambda, t)$ are interference independent of the true signal R_{real}, we combine them as $M(\lambda, r, t)$.

Simplifying (1), R_{show} is composed of two primary elements: $R_{\text{real}}(\lambda, t) \cdot O(\lambda, r) \cdot \tau(\lambda, d, w)$ and $M(\lambda, r, t)$. Our goal is to improve the signal-to-noise ratio (SNR), as detailed in (2). This involves recovering the signal R_{real} from the low-frequency degradation caused by $O(\lambda, r)$ and $\tau(\lambda, d, w)$, while simultaneously weakening the interference $M(\lambda, r, t)$. This dual-tuning approach forms the foundation of the DTNet design. In Sections III-B–III-D, we will provide detailed descriptions of our designs

$$\text{SNR} = \frac{R_{\text{real}}(\lambda, t) \cdot O(\lambda, r) \cdot \tau(\lambda, d, w)}{M(\lambda, r, t)}. \tag{2}$$

B. Dynamic Guided Filtering Module

$M(\lambda, r, t)$, which superimposes various types of noise and stray radiation, has a wide band in the spectrum. Because of precious details in infrared images, $M(\lambda, r, t)$ is more harmful to the high-frequency components than to the low-frequency parts. Thus, the DGFM is developed to remove the high-frequency part of $M(\lambda, r, t)$ while preserving the details in infrared images.

Guided filter [71] is an edge-preserving smoothing approach, whose filtered output is achieved through the local linear transformation of the guide image. The formula is shown as follows:

$$O_i = a_k \cdot G_i + b_k \quad \forall i \in \omega_k \tag{3}$$

where G is the guide image, O is the filtered output, a_k and b_k are linear coefficients which are constant for all pixels in a window. ω_k denotes the window which is a small area centered at pixel i.

Given the original image I, solve for a_k and b_k by minimizing the difference between the output image O and I within

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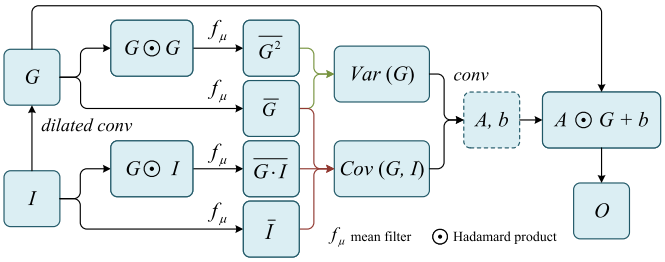


Fig. 1. Illustration of DGFM. The feature maps I first pass through a dilated convolution (dilated conv) to generate the guide maps G. Subsequently, the window variance of the guide maps $\mathrm{Var}(G)$ and the window covariance of the guide maps with the original feature maps $\mathrm{Cov}(G, I)$ are calculated separately. Next, $\mathrm{Var}(G)$ and $\mathrm{Cov}(G, I)$ are concatenated and used to generate the parameter A via convolution (conv). Finally, the guide maps G are linearly combined to produce the filtered result O.

local windows. The formula for this process is presented below

$$\min \sum_{i \in \omega_k} ((a_k \cdot G_i + b_k - I_i)^2 + \varepsilon \cdot a_k^2) \tag{4}$$

$$a_k = \frac{\mathrm{Cov}(G, I)^{\omega_k}}{\mathrm{Var}(G)^{\omega_k} + \varepsilon} \tag{5}$$

$$b_k = \bar{I}_k - a_k \cdot \bar{G}_k \tag{6}$$

where $\mathrm{Cov}(G, I)^{\omega_k}$ denotes the covariance between the guide image G and the original image I in window ω_k, $\mathrm{Var}(G)^{\omega_k}$ stands for the variance of the guide image G in window ω_k, and ε is a constant which prevents the denominator from being 0. $\bar{I}_k$ and $\bar{G}_k$ represent the mean values of the original image I and guide image G within the window ω_k, respectively.

Since the pixel at position i exists in multiple surrounding windows simultaneously, the average value from these windows is adopted to represent it. The formula is given below

$$O_i = \frac{1}{|\Omega_i|} \sum_{k \in \Omega_i} (a_k \cdot G_i + b_k) \tag{7}$$

where Ω_i is the set of windows containing pixel i, and $|\Omega_i|$ is the number of windows in set Ω_i.

However, there are two problems in the guided filter: 1) a hyperparameter ε needs to be specified in (5), which significantly affects the performance of edge-preserving denoising and 2) it is not optimal to combine the values of pixel i equally in each window as described in (7), as this may limit the effectiveness of the guided filter. Therefore, we employ learnable parameters to fit ε and $1/|\Omega_i|$ in our DGFM, which produces the optimal value via the supervision of oriented bounding boxes. Furthermore, the guide image G is the same as the original image I in the initial guide filter. According to (3), the filtered output O has the same gradient as the guide image G, which means that the edge features in the guide image directly contribute to the filtering result. Thus, we generate the guide image by applying a dilation convolution to the original image in DGFM, which is proven to enhance edge features [9]. The detailed structure of DGFM is presented in Fig. 1. We achieve the optimization of $1/|\Omega_i|$ and ε through the weights and biases of a convolution operator.

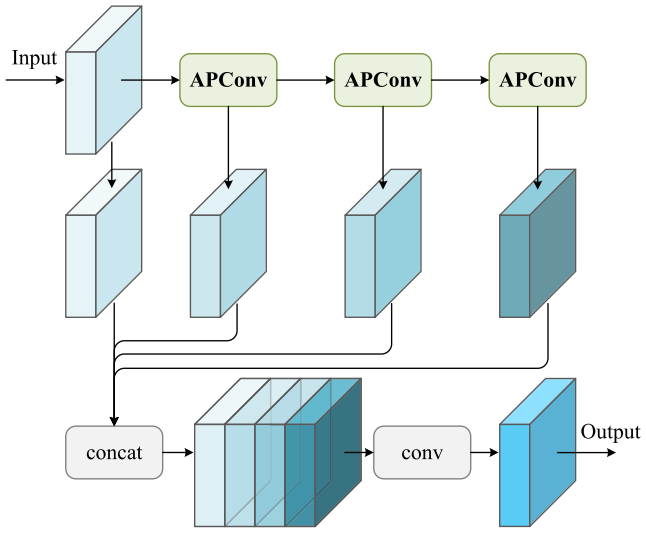


Fig. 2. Illustration presents the PSRM, with APConv serving as its core component. "conv" and "concate" are abbreviations for "1 × 1 convolution" and "concatenate," respectively.

C. Point Spread Recovery Module

After applying the dynamic guided filter module (DGFM) to remove high-frequency interference, the infrared image retains the components: $R_{\mathrm{real}}(\lambda, t) \cdot O(\lambda, r) \cdot \tau(\lambda, d, w)$ and the low-frequency portion of $M(\lambda, r, t)$. Both $O(\lambda, r)$ and $\tau(\lambda, d, w)$ disperse the radiant energy from individual pixels to their neighbors, diminishing signal intensity. The low-frequency of the $M(\lambda, r, t)$ elevates disturbance levels. In response, we propose the PSRM, which is designed for adaptive energy aggregation to focus scattered energy and suppress interference, enhancing the SNR.

1) PSRM Structure: The structure of the PSRM is shown in Fig. 2. In the PSRM, feature maps are sequentially processed through a series of APConvs. The APConv, as the core component of PSRM, serves as the feature extraction operator designed based on the mechanism of low-frequency degradation. While a single APConv is theoretically capable of alleviating multiple degradations simultaneously, using a series of APConvs in a cooperative manner enables each one to concentrate on recovering specific parts of degradation. The ideal number of APConv will be determined experimentally in Section IV-E. The feature maps produced by each APConv are subsequently concatenated, and their channels are aligned with the input through a 1×1 convolution. The following paragraph will elaborate on the design concept of APConv.

2) Adaptive Pyramidal Convolution: According to the Fraunhofer diffraction theory [70], the point spread function of diffraction $O(\lambda, r)$ can be expressed as the square of the first-order Bessel function. The formula and shape of the $O(\lambda, r)$ are shown in the following equation and Fig. 3(a), respectively,

$$O(\lambda, r) = \left(\frac{2J_1\left(\frac{2\pi r\sqrt{x^2+y^2}}{\lambda}\right)}{\frac{2\pi r\sqrt{x^2+y^2}}{\lambda}} \right)^2. \tag{8}$$

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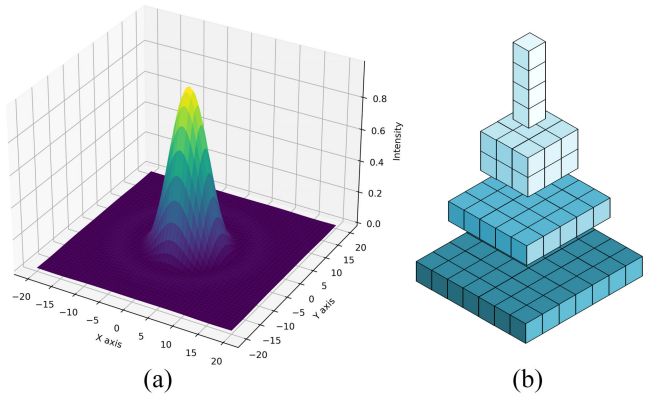


Fig. 3. Diagrams of the diffraction function and the pyramid kernel are presented. (a) Displays the image of the point spread function caused by diffraction, while (b) shows the pyramid-shaped kernel used to aggregate dispersed energy.

J_1 is the first-order Bessel function. λ and r are the wavelength and radius of the aperture, respectively. x and y are positions in a coordinate system whose origin is the center of the aperture.

As shown in Fig. 3(a), diffraction spreads a small portion of the energy to the surroundings, yet the majority of the energy remains concentrated at the center [10]. Therefore, we employ kernels shaped like a pyramid, known as APConv, to aggregate dispersed energy. This is illustrated in Fig. 3(b). The pyramidal configuration of the kernels provides different sizes of receptive fields for various ranges of energy aggregation. Moreover, to better align with the distribution of the diffraction function $O(\lambda, r)$, we allocate a larger proportion of small receptive field kernels and a smaller proportion of large receptive field kernels. This means that more channels in the feature maps focus on the dominant part of the energy, while fewer channels are dedicated to the dispersed part.

The atmospheric degradation model $\tau(\lambda, d, w)$ is more complex than diffraction. Long-wave infrared undergoes absorption, re-radiation, and scattering during propagation through the atmosphere, which is affected by weather and imaging distance. Atmospheric absorption leads to the attenuation of radiant energy, while re-radiation and scattering cause the energy from the center to disperse into the surroundings. The absorption and re-radiation of long-wave infrared by atmospheric particles considerably surpasses the effect of scattering [12], [13]. According to Planck's law [69], thermal radiation from an object is isotropic. Thus, the model for atmospheric attenuation can be approximated by a 2-D Gaussian-like function, which can also be effectively fit using APConv. Since $\tau(\lambda, d, w)$ varies with weather and distance, it may differ for different images. Hence, we adopt an adaptive weight generation strategy in APConv, which generates suitable recovery operators based on the specific conditions present in the infrared image.

For removing the low-frequency portion of $M(\lambda, r, t)$, we employ a global-guided strategy in the adaptive weight generation. Because the feature maps processed by APConv are decomposed by the filters with different cutoff frequencies in DTBlock which will be described in Section III-D. By subtracting the outputs of low-pass feature maps with different cutoff frequencies, we can create bandpass feature maps that are equivalent to the desired frequency band. The global weights contribute to achieving signal enhancement and interference suppression in feature maps with different frequencies.

Algorithm 1 Pseudocode of APConv in a PyTorch-Like Style

```
# Input: x: [B, C, H, W], Output: x_out: [B, C, H, W]
# B: batch size, H: height, W: width, C: channel number
# k: list of kernel sizes, d: list of kernel dilation
# p: list of kernel padding, n: list of kernel numbers
# sum(n) = C, len(k) = len(d) = len(p)
# # # # # # # # # # # initialization # # # # # # # # # # #
convs = nn.ModuleList()
for i in range(len(k)):
    convs.append(nn.Conv2d(n[i], n[i], k[i], d[i], p[i]))
num_k = [nk ** 2 for nk in k]
mlp = nn.Conv2d(C, sum(num_k), k=1, s=1)
avg_pool = nn.AdaptiveAvgPool2d(1)
fc = nn.Linear(C, C)
unfolds = nn.ModuleList()
for i in range(len(k)):
    unfolds.append(nn.Unfold(k[i], d[i], p[i], s=1))
# # # # # # # # # # # # forward # # # # # # # # # # # # #
x_split = torch.split(x, n, dim=1)
x_hat = []
for i in range(len(k)):
    x_hat.append(convs[i](x_split[i]))
x_hat= torch.cat(x_hat, dim=1)
# weight: [B, num_k, H, W] -> [B, 1, num_k, H, W]
weight = mlp(x_hat).unsqueeze(1)
weights = torch.split(weight, num_k, dim=2)
global = fc(ave_pool(x)).view(B, C, 1, 1, 1)
globals = torch.split(global, n, dim=1)
# broadcast multiplication
weights = [w * g for w, g in zip(weights, globals)]
x_out = []
for i in range(len(k)):
    x_unfold = unfolds[i](x_split[i])
    x_unfold = (weights[i] * x_unfold).sum(dim=2)
    x_out.append(x_unfold.view(B, n[i], H, W))
x_out = torch.cat(x_out, dim=1)
return x_out
```

The detailed procedure of APConv is shown in Fig. 4 and pseudocode Algorithm 1. First, feature maps are fused by kernels with different receptive fields in space and by a 1×1 convolution layer in channels. Then, the number of channels is adjusted to match the sum of elements of all kernels using a 1×1 convolution. The channels of the fused feature maps are split according to the number of elements in each kernel. The initial feature maps pass through global average pooling and a linear layer to generate global-guided factors. Finally, the adaptive pyramid kernels are generated by broadcasting multiplication between the fused feature maps and global-guided factors.

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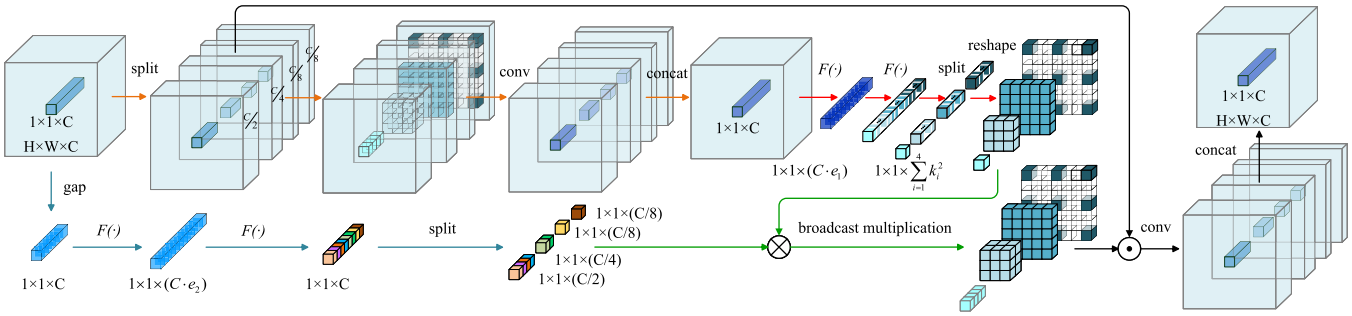


Fig. 4. Operation of adaptive pyramid convolution (APConv) is depicted using five colored arrows, each representing a different stage in the process. The orange arrow denotes the feature fusion stage preceding the generation of the adaptive pyramid kernel. The red arrow depicts the formation of the APConv prototype, which has only one kernel at each level of the pyramid. The blue arrow depicts the process of generating the global-guided factor. The green arrow represents the broadcasting of the global-guide factor onto the APConv prototype, resulting in its element-wise multiplication to form the fully operational APConv. Finally, the black arrow represents the convolution of APConv with the feature maps to produce the output. "conv" and "concate" are abbreviations for "1 × 1 convolution" and "concatenate," respectively.

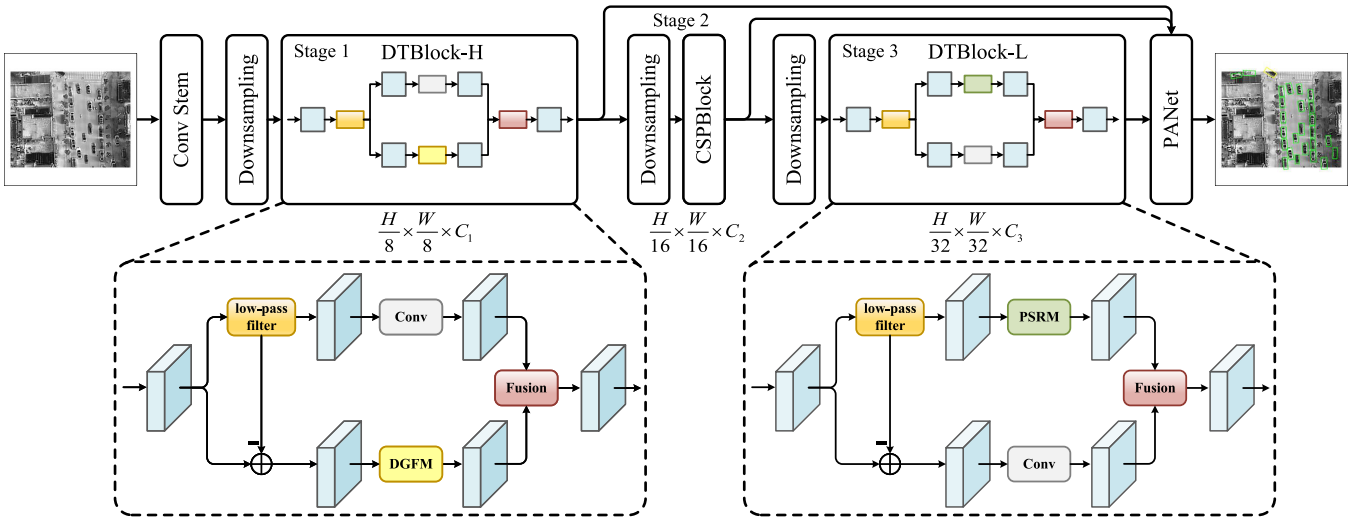


Fig. 5. Framework of the proposed DTNet for infrared images detection. It consists of five main components: Conv Stem, Downsampling, DTBlock, CSPBlock, and PANet. The Conv Stem, consisting of two CSPBlocks followed by Downsampling, serves as the initial processing unit and outputs 4× downsampled feature maps. The CSPBlock is a lightweight structure with excellent feature extraction performance. PANet is designed for fusing features from various layers. The DTBlock is available in two forms, DTBlock-H and DTBlock-L, which combine DGFM and PSRM, respectively. The orange and red parts in DTBlock represent the feature map frequency decomposition and fusion operations, respectively. "Conv" is an abbreviation for "1 × 1 convolution."

D. Dual-Tuning Network

For better signal reinforcement and interference suppression, the DTBlock is adopted to complement the PSRM and DGFM. In the DTBlock, feature maps are first decomposed into high- and low-frequency parts by low-pass filters with learnable cutoff frequencies. Then, different modules are applied to process the feature maps with varying frequencies. Finally, the two parts are suitably fused. We use DGFM to adjust the high-frequency features and PSRM to optimize the low-frequency features in the DTBlock. The introduction of DTBlock enables DGFM to avoid smoothing the low-frequency content of infrared images and allows PSRM to recover low-frequency degradation without interference from high-frequency noise. Furthermore, the low-pass or high-pass feature maps, decomposed by learnable low-pass filters, may combine to produce band-pass feature maps. These are favorable for the integration of diffuse energy and the delicate removal of wide-band interference.

The architecture of our DTNet, as illustrated in Fig. 5, consists of five components: Conv Stem, Downsampling, DTBlock, CSPBlock [72], and PANet [73]. The CSPBlock, recognized for its lightweight structure, offers exceptional feature extraction performance, enhancing the efficiency of the network. The Conv Stem, comprising two CSPBlocks, performs a twofold downsampling of the input image through the convolution with the stride of 2. This is followed by a Downsampling process, which achieves an additional twofold downsampling via maximum pooling with a stride of 2. The PANet stands out for its ability to fuse features across different layers, which integrates multiscale feature maps from stage 1–3. The DTBlock is differentiated into two variants: DTBlock-H and DTBlock-L, which are designed to work with DGFM and PSRM, respectively. Both DTBlock-H and DTBlock-L serve to decompose feature maps into high- and low-frequency components by applying specific filters. In DTBlock-H, the DGFM specifically targets the high-frequency part of feature maps to suppress interference and amplify the edges of targets. Conversely, in DTBlock-L,

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the PSRM focuses on aggregating radiant energy dispersed across the low-frequency part of feature maps. The structure and fusion of the DTBlock will be discussed in Section IV.

IV. Experiments

In this section, we compare our DTNet with several SOTA approaches on oriented bounding box detection and vehicle detection by quantitative and qualitative comparisons. We first describe the configuration and implementation details of experiments, including the dataset, the evaluation metrics, the network, and the optimizer. Subsequently, we conduct numerous ablation experiments with SOTA alternatives to demonstrate the effectiveness of our designs. Then, we investigate the effect of parameters on the features and results of the network. Finally, we analyze and explain the improvements with visualization.

A. Experimental Configurations

We verify our approach on the DroneVehicle [15] which is a publicly available aerial image dataset of oriented vehicles. The details of datasets and evaluation metrics are as follows. DroneVehicle is the largest RGB-Infrared cross-modality vehicle detection dataset on aerial images. It includes 28 439 pairs of infrared and visible images with oriented bounding box annotations of 953 087 instances. Specifically, it is divided into three parts: training, validation, and test set, with 17 990, 1469, and 8980 images, respectively. The scale of images is 840 $\times$ 712 pixels, which extends 100 pixels of a white border in the original 640 $\times$ 512 size. In addition, DroneVehicle stands as the most comprehensive infrared vehicle detection dataset currently available, offering annotations for five categories: car, truck, bus, van, and freight car. It spans multiple scenarios, including urban roads, residential areas, and parking lots, and encompasses a diverse array of imaging conditions, such as various time periods, atmospheric conditions, and imaging distances, making it a robust resource for infrared vehicle detection and analysis. In this article, we only adopt infrared images of the dataset and cut them into a square of 704 $\times$ 704 for operational purposes. Since the removed part is the white border, it does not affect the experimental results.

We selected three indicators to quantify the approach, including the mean average precision (mAP), the number of parameters (Params), and the floating-point operations (FLOPs). The mAP is to evaluate the effectiveness of the model. We follow the mAP@0.5 and mAP@.5:.95 of COCO protocol. The Params and FLOPs are used to measure the capacity of the model for a more comprehensive evaluation.

B. Implementation Details

We set four types of kernels in the APConv whose kernel sizes and dilation rates are 1, 3, 5, 3 and 1, 1, 1, 3, respectively. The ratios for these kernels are set at 4:2:1:1. The learnable frequency decomposition filters use Gaussian-like kernels, which is the orange part in DTBlock. The spatial attention fusion approach is employed, indicated by the red part in DTBlock. There are three APConvs in PSRM when DTNet is the base form (DTNet-B). Besides, we offer two other versions to cater to different requirements: a smaller version (DTNet-S) and a tiny version (DTNet-T). The depth and width of the models for DTNet-S and DTNet-T are, respectively, 2/3 and 1/3 of those for DTNet-B. It has been experimentally proven that the network outputs the best result with these settings. We will discuss the impact of these settings on network performance in detail in Sections IV-D and IV-E. The batch size is set to 32 in the dataloader. During the training stage, we perform image augmentations such as rotation and mosaic. The parameters of our DTNet are optimized using stochastic gradient descent (SGD) with momentum. The momentum is set to 0.93, and the weight decay is set to 5×10^{-4}. The learning rate is initially set to 5×10^{-2} and then exponentially decayed from epoch 0 to 200. The comparative experiments in this article are implemented on the MMRotate platform [83]. All experiments are conducted using the PyTorch framework [84] on an NVIDIA GeForce RTX 3090.

C. Comparison to SOTA

In this section, we first compare our DTNet against various SOTA methods on oriented bounding box detection and vehicle detection described in Section II-A. We primarily focus on methods designed for detecting densely packed and arbitrarily oriented vehicles in aerial imagery. Subsequently, we substitute the backbones of our DTNet with numerous SOTA networks, demonstrating the enhanced feature extraction capabilities of our approach for infrared images. All experiments are conducted using the DroneVehicle dataset.

1) Comparison to SOTA Frameworks: The results of comparisons between DTNet and SOTA frameworks in visible and infrared vehicle detection, as presented in Table I, demonstrate that DTNet outperforms others in terms of mAP@.5 and mAP@.5:.95. It also secures three top scores and one second-best score in five vehicle subcategories. Moreover, our approach is notably competitive regarding the number of parameters and floating-point operations, boasting the smallest Params and the third smallest FLOPs. Most of the algorithms included in our comparison are ranked on the DOTA leaderboard. Compared to I^2MDet, which previously achieved the best performance on the infrared vehicle detection in the DroneVehicle, DTNet shows an improvement of 0.93 and 6.65 in mAP@.5 and mAP@.5:.95, respectively. Furthermore, our approach shows a significant advantage in single-modality infrared detection with a 14.23 increase in mAP@.5 over UA-CMDet, which is a multimodal infrared-visible detection method for DroneVehicle.

2) Comparison to SOTA Backbones: Besides the visible and infrared vehicle detection frameworks discussed earlier, we further benchmark our DTNet against the advanced feature extraction networks outlined in Section II-C. To cater to a wide spectrum of requirements, we introduce three variants of DTNet: DTNet-T (tiny version), DTNet-S (small version), and DTNet-B (base version). Table II presents the experimental comparisons between various SOTA networks, including those based on convolutional, attention, and hybrid architectures. LSKNet, noted as the SOTA on the DOTA dataset as of July 2023, is included in the comparison

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TABLE I

RESULTS FROM COMPARATIVE EXPERIMENTS AGAINST VARIOUS SOTA METHODS IN VISIBLE AND INFRARED VEHICLE DETECTION. THE EVALUATION METRIC FOR THE VEHICLE SUBCATEGORY IS AP@.5. WE ADOPT MAP@.5 AND MAP@.5:.95 IN THE COCO PROTOCOL AS EVALUATION METRICS OF THE MEAN AVERAGE PRECISION. PARAMS AND FLOPS ARE THE NUMBER OF PARAMETERS AND THE FLOATING-POINT OPERATIONS, RESPECTIVELY, TO MEASURE THE CAPACITY OF THE MODEL. **RED** INDICATES THE BEST RESULT. **BLUE** INDICATES THE SECOND-BEST RESULT

Methods	car	truck	van	bus	freight	mAP@.5	mAP@.5:.95	Params.(M)	FLOPs(G)
Rotated RetinaNet [74]	88.3	20.7	19.6	66.2	32.5	45.47	27.76	36.21	99.95
Rotated Faster R-CNN [75]	89.9	48.9	45.8	86.9	48.3	63.67	36.04	41.13	107.20
Rotated RepPoints [76]	89.4	50.4	42.9	72.2	42.1	59.39	35.82	36.61	91.87
Rotated FCOS [77]	89.8	48.6	43.4	88.5	50.2	64.08	40.07	31.90	97.59
Rotated ATSS [78]	89.6	38.2	26.5	84.9	28.2	53.48	32.52	36.02	98.04
RoI Transformer [39]	90.1	60.4	52.2	89.7	58.9	70.29	43.57	55.06	121.13
Gliding Vertex [79]	89.9	47.0	44.4	88.1	49.1	62.96	35.04	41.13	107.20
CSL [26]	88.4	22.4	22.5	60.0	32.1	44.96	26.77	37.14	109.58
R^3Det [29]	89.8	48.2	44.5	88.1	49.6	63.75	36.77	36.95	107.63
UA-CMDet [15]	87.5	60.7	38.0	87.1	46.8	64.00	-	-	-
S^2ANet [41]	89.8	45.2	39.4	88.8	44.6	61.52	35.30	38.55	92.96
ReDet [80]	**90.2**	57.6	48.3	89.7	54.5	68.09	42.44	**31.57**	**52.56**
CFA [35]	90.0	58.7	48.0	88.5	51.7	67.31	42.59	36.60	91.85
Oriented R-CNN [40]	89.9	56.6	46.9	89.6	54.4	67.52	42.60	41.13	107.26
GWD [31]	86.8	24.8	13.8	72.3	28.0	45.04	26.02	36.21	99.95
KLD [32]	89.9	39.7	33.7	87.0	35.5	57.17	34.58	41.67	156.37
SASM [43]	89.7	51.9	38.0	86.2	49.8	62.88	40.65	36.60	91.85
Oriented RepPoints [36]	89.9	54.8	47.5	88.4	52.2	66.98	41.32	36.60	91.85
KFIoU [33]	90.1	56.2	51.7	**89.8**	52.3	68.00	40.22	55.06	121.13
PSC [30]	89.9	52.2	47.9	88.9	53.4	66.40	41.71	31.91	97.71
ViT-B+RVSA [81]	89.7	52.3	44.4	88.0	51.0	65.07	42.63	114.37	413.60
I^2MDet [9]	**96.3**	**73.4**	**58.6**	**93.2**	**65.0**	**77.30**	**46.20**	31.87	**48.90**
DTNet-B (ours)	**90.2**	**78.1**	**65.7**	89.2	**67.9**	**78.23**	**52.85**	**27.67**	57.10

TABLE II

RESULTS FROM COMPARATIVE EXPERIMENTS AGAINST VARIOUS SOTA NETWORKS OF FEATURE EXTRACTION. TYPES C, A, AND H CORRESPOND TO CONVOLUTIONAL, ATTENTIONAL, AND HYBRID ARCHITECTURES, RESPECTIVELY

Backbone	Type	Param. (M)	FLOPs (G)	mAP @.5	mAP @.5:.95
MobileNet V3-S/050 [82]	C	4.01	7.5	68.33	41.16
EfficientViT-M0 [65]	A	4.37	42.7	70.69	43.25
FasterNet-T0 [64]	C	6.04	12.8	73.86	47.68
EMO-5M [60]	H	8.60	104.1	74.32	48.08
DTNet-T (ours)	C	2.21	5.4	75.04	49.41
PoolFormer-S24 [56]	A	25.03	63.1	61.37	37.07
UniRepLKNet-F [55]	C	10.03	23.3	68.09	42.78
SKResNet-18 [49]	C	15.58	37.6	71.70	46.38
FasterNet-T1 [64]	C	9.80	21.7	74.09	47.01
LSKNet-S [54]	C	15.88	43.9	74.34	49.31
DTNet-S (ours)	C	9.60	20.5	76.18	50.16
PoolFormer-S36 [56]	A	34.50	88.9	63.49	38.16
InceptionNeXt-T [51]	C	28.58	76.7	65.06	39.98
ConvNeXt V2-T [50]	C	33.24	80.9	70.41	43.66
UniRepLKNet-T [55]	C	35.41	91.6	71.16	44.38
Swin-T [53]	A	32.12	404.6	71.94	45.02
SKResNet-50 [49]	C	31.68	78.8	72.91	47.06
NextViT-S [61]	H	29.35	83.2	74.75	48.83
ResNet-50S [47]	C	31.55	101.5	74.78	48.08
FasterNet-S [64]	C	33.73	83.4	75.07	48.70
I^2MDet [9]	H	29.14	46.4	76.02	45.15
DTNet-B (ours)	C	27.67	57.10	78.23	52.85

alongside other networks that have achieved SOTA status in tasks like recognition, detection, and semantic segmentation. Significantly, the DTNet family outperforms these models by requiring the fewest parameters and floating-point operations, highlighting its exceptional capabilities in infrared vehicle detection.

TABLE III

ABLATION STUDY OF DTNET STRUCTURE. WE ADOPT MAP@.5 AND MAP@.5:.95 IN THE COCO PROTOCOL AS EVALUATION METRICS OF THE MEAN AVERAGE PRECISION. BOLDED INDICATES THE BEST RESULT

DTNet-B		mAP@.5	mAP@.5:.95
Stage 1	Stage 3		
CSPBlock	CSPBlock	74.57	49.01
DTBlock-A	DTBlock-A	76.68	51.97
DTBlock-B	DTBlock-B	77.13	52.17
DTBlock-H	DTBlock-L	77.93	52.20
DTBlock-L	DTBlock-H	**78.23**	**52.85**

D. Ablation Study

In the following discussion, we conduct extensive ablation studies to verify the effectiveness of our approach. This includes investigating the optimal structure of DTNet, identifying the ideal components in DTBlock, and evaluating the performance of APConv in PSRM.

1) Structure of DTNet: We employ DTNet-B as the foundational model to investigate the impact of its structural variations on performance, as detailed in Table III. The baseline model, which substitutes DTBlocks in DTNet-B with CSPBlocks in the third and fifth stages, is presented in the first row of Table III. We specifically developed DTBlock for vehicle detection in infrared images, with the central principle being the frequency decomposition of feature maps and the subsequent separate processing of the low- and high-frequency components. As illustrated in Fig. 6, we designed four DTBlock structural variations. Both DTBlock-A and DTBlock-B incorporate PSRM and DGFM within the same

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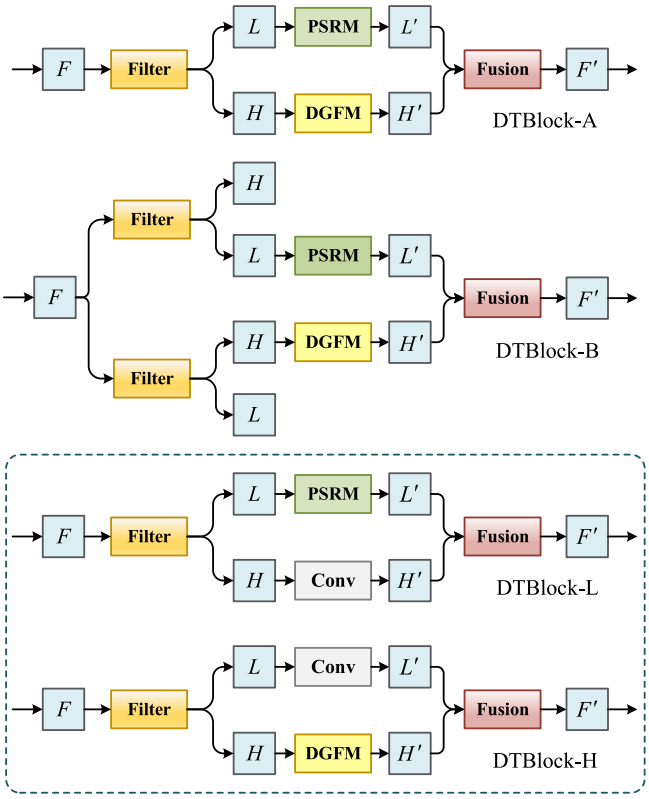


Fig. 6. Illustration shows different DTBlock variations. DTBlock-H and DTBlock-L should be used together. The *Filter* block serves to decompose the input feature map F, into its low-frequency and high-frequency parts, denoted as L and H, respectively. The symbols L' and H' represent the respective components post-processing. These processed components, L' and H', are subsequently integrated by the *Fusion* block to produce the output feature map F'.

DTBlock. The difference primarily lies in the approach to frequency-domain decomposition: DTBlock-A employs a single set of low-pass filters, whereas DTBlock-B utilizes two sets. This design consideration stems from our understanding that the feature maps decomposed by a single filter set are complementary in the frequency domain, but the optimal frequency bands of features for PSRM and DGFM processing may overlap, potentially leading to conflicts during filter optimization. The comparative results detailed in the second and third rows of Table III affirm the advantage of DTBlock-B over DTBlock-A. However, there are more parameters to be optimized and are more computationally intensive in DTBlock-B. Thus, we separate PSRM and DGFM into two DTBlocks, designated as DTBlock-L and DTBlock-H, to optimize their performance and ensure computational efficiency. The fourth and fifth of Table III illustrate the difference in performance of DTBlock-L and DTBlock-H when positioned at distinct locations in the backbone. The results in Table III highlight the performance enhancements afforded by DTBlock and discuss the advantageous configurations of both DTNet and DTBlock.

2) Components in DTBlock: Two key components of DTBlock are the frequency-domain decomposition and the feature maps fusion components, denoted as *Filter* and *Fusion* in Fig. 6. The impact of employing different frequency decomposition strategies in DTBlock-L and DTBlock-H is

TABLE IV

ABLATION STUDY ON THE FREQUENCY DECOMPOSITION STRATEGY IN DTBLOCK. WE ADOPT mAP@.5 AND mAP@.5:.95 IN THE COCO PROTOCOL AS EVALUATION METRICS OF THE MEAN AVERAGE PRECISION. BOLDED INDICATES THE BEST RESULT

Frequency Decomposition Strategy	mAP@.5	mAP@.5:.95
No Decomposition	77.54	51.64
Learnable Bandpass Filters	77.89	52.16
Learnable Low-pass Filters	**78.23**	**52.85**

TABLE V

ABLATION STUDY ON THE FUSION STRATEGY IN DTBLOCK. WE ADOPT mAP@.5 AND mAP@.5:.95 IN THE COCO PROTOCOL AS EVALUATION METRICS OF THE MEAN AVERAGE PRECISION. BOLDED INDICATES THE BEST RESULT

Fusion Strategy	mAP@.5	mAP@.5:.95
Addition	77.11	50.02
Concatenate	76.73	49.26
Channel Attention	76.60	50.19
Frequency Ratio	77.47	51.22
Spatial Attention	**78.23**	**52.85**

detailed in Table IV. The first row of Table IV presents the performance without the decomposition strategy, which demonstrates the enhancement from dual tuning compared to the second and third rows. We explore two decomposition strategies: Learnable Bandpass Filters and Learnable Low-pass Filters. Specifically, for the Learnable Bandpass Filters, we transform feature maps into the frequency domain via Fourier transform and then apply learnable masks to the spectrogram for bandpass filtering. The equations for the Learnable Bandpass Filters are as follows:

$$\text{mask} = X[\mathscr{F}(\text{feat map})] \tag{9}$$

$$(\text{feat map})_L = \mathscr{F}^{-1}[\mathscr{F}(\text{feat map}) \odot \text{mask}] \tag{10}$$

$$(\text{feat map})_H = \text{feate map} - (\text{feat map})_L \tag{11}$$

where $\mathscr{F}$ and $\mathscr{F}^{-1}$ represent the Fourier transform and the inverse Fourier transform, respectively, X is typically a transformation function, often a 2-D convolution. The symbol $\odot$ denotes the Hadamard product. $(\text{feat map})_L$ and $(\text{feat map})_H$ refer to the decomposed feature maps.

For the Learnable Low-pass Filters, we initialize the convolution kernel with a discrete 2-D Gaussian function, followed by updates through optimization. Parameters of the kernel are then mapped using a softmax function to maintain low-pass characteristics. The equations for the Learnable Bandpass Filters are as follows:

$$\text{kernel}_{\text{low-pass}} = \text{softmax}(\text{kernel}) \tag{12}$$

$$(\text{feat map})_L = \text{feat map} * \text{kernel}_{\text{low-pass}} \tag{13}$$

$$(\text{feat map})_H = \text{feat map} - (\text{feat map})_L \tag{14}$$

where $\text{kernel}_{\text{low-pass}}$ represents the low-pass filter and the symbol $*$ indicates convolution.

Findings in the last two rows of Table IV demonstrate that incorporating prior knowledge of low-pass decomposition benefits performance. This confirmation also supports the

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TABLE VI

ABLATION STUDY ON APCONV IN PSRM. EVALUATING WITH mAP@.5, mAP@.5:.95 (COCO PROTOCOL), PARAMS, AND FLOPS. BOLDED INDICATES THE BEST RESULT

PSRM	mAP@.5	mAP@.5:.95	Param.(M)	FLOPs(G)
ODConv [57]	74.29	48.39	39.10	44.0
Involution [59]	76.52	51.46	24.61	57.0
AWD [86]	76.54	51.41	27.05	59.0
PyConv [87]	73.52	46.30	25.11	52.3
APConv (ours)	**78.23**	**52.85**	27.67	57.1

TABLE VII

EFFECT OF THE NUMBER OF APCONVS IN THE PSRM IS STUDIED. EVALUATING WITH mAP@.5, mAP@.5:.95 (COCO PROTOCOL), PARAMS, AND FLOPS

Number of APConvs	mAP @.5	mAP @.5:.95	Params. (M)	FLOPs (G)
1	76.82	51.50	21.88	52.5
2	77.97	52.39	24.78	54.8
3	78.23	52.85	27.67	57.1
4	78.27	53.01	30.55	59.4

effectiveness of PSRM and DGFM in extracting low-frequency and high-frequency features, respectively.

We investigate five options for the *Fusion* component in DTBlock: addition, concatenation, channel attention, frequency ratio weighting, and spatial attention. Addition and concatenation are conventional methods for fusing feature maps in networks. Frequency ratio weighting is a fusion strategy that utilizes the ratio between the low- and high-frequency components of the feature maps as fusion weights. The channel attention fusion approach, inspired by squeeze-and-excitation networks (SENet) [85], leverages channel-wise feature weights for the low- and high-frequency parts. Spatial attention predicts a mask of the same size as the feature map to effectively merge low- and high-frequency components. Their experimental results are presented in Table V, where spatial attention achieved the best performance.

3) Ablation on APConv: APConv, a feature extraction operator applied in PSRM, is designed according to the degradation mechanisms of infrared images. It is worth noting that there are several operators that have been proposed that are similar to our APConv, while they are essentially different. omni-dimensional dynamic convolution (ODConv) [57], adaptive weighted downsampling (AWD) [86], and Involution [59] are dynamic convolutions like our APConv with input-dependent kernels. However, these operators maintain fixed receptive fields for their kernels, treating diverse ranges of information uniformly. The pyramid-shaped operators [48], [87], [88], [89], as represented by PyConv [87], employ kernels with various receptive fields, similar to APConv. Despite this, kernels with different receptive fields process the same ratio of feature maps in them, and there is no implementation of the adaptive weight generation strategy. The evidence in Table VI indicates the advantages of APConv for infrared vehicle detection, when the number of parameters and floating-point operations are comparable to other similar operators.

TABLE VIII

EFFECT OF SIZE, DILATION RATE, AND QUANTITY RATIO OF THE KERNEL IN APCONV IS STUDIED. WE ADOPT mAP@.5 AND mAP@.5:.95 IN THE COCO PROTOCOL AS EVALUATION METRICS. BOLDED INDICATES THE BEST RESULT

APConv			mAP@.5	mAP@.5:.95
kernel sizes	dilation rates	ratios		
1	1	1	77.76	51.98
3	3	1	77.82	52.39
1, 3, 3, 3	1, 1, 2, 3	1: 1: 1: 1	77.87	52.60
1, 3, 3, 3	1, 1, 2, 3	4: 2: 1: 1	78.08	52.73
1, 3, 5, 3	1, 1, 1, 3	4: 2: 1: 1	**78.23**	**52.85**
1, 3, 5, 3	1, 1, 2, 5	4: 2: 1: 1	77.70	52.03
1, 3, 5, 3	1, 1, 3, 8	4: 2: 1: 1	77.84	52.14

E. Parameter Analysis

In this section, we investigate the impact of varying the number of APConvs in PSRM and the parameters within APConvs.

1) Parameter Analysis on PSRM: The role of APConv in mitigating various degradations during infrared imaging has been previously discussed in Section III-C. While a single APConv theoretically possesses the capability to recover from diverse degradations, increasing the number of APConvs in PSRM can enhance robustness and simplify optimization. Experimental results in Table VII indicate that network capacity incrementally improves with an increasing number of APConvs. However, performance gains nearly plateau when escalating from 3 to 4 APConvs. Consequently, in balancing performance enhancement and computational complexity, the number of APConvs in DTNet-B is set to 3. For the DTNet-T and DTNet-S configurations, the number of APConvs is set to 1 and 2, respectively, aligning with their intended operational scopes and computational constraints.

2) Parameter Analysis on APConv: This part determines the optimal parameters for APConv, including kernel sizes, dilation rates, and the ratios between kernels of various receptive fields. The experimental findings are summarized in Table VIII. The first and second rows of Table VIII indicate the performance of 1×1 and 3×3 kernels employed in APConv, serving as benchmarks for the assessment of kernel parameters in APConv. In the third row of Table VIII, we set the dilation rates for the 3×3 kernels, which are the prototype of APConv. Compared to the initial experiments, the pyramid-shaped kernels in the third row achieve enhanced performance without an increase in computational complexity. Then, according to the infrared energy diffusion mechanism, the kernel with a small receptive field in the pyramidal structure is set to a larger ratio, whose result is displayed in the fourth row of Table VIII. The experiments in the first four rows validate our concept of designing pyramidal kernels to extract features based on point spread functions. In other words, a higher proportion of small receptive field kernels is employed to extract features from the central region, while a few large receptive field kernels aggregate dispersed energy. Finally, spanning from

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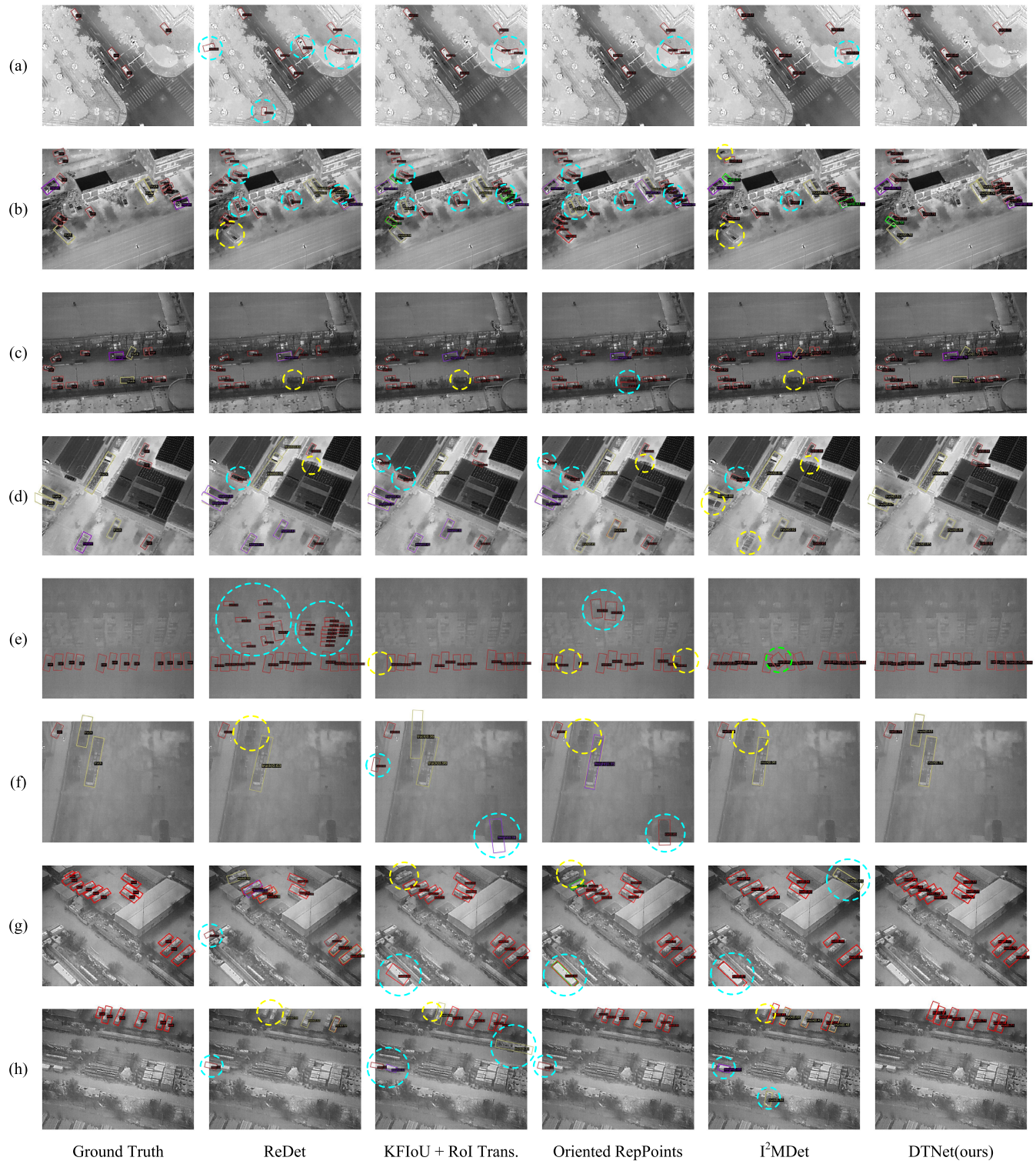


Fig. 7. Illustration of output visualization comparing detection results from ReDet, RoI Transformer with KFIoU, Oriented RepPoints, and I^{2}MDet. We compare the differences between these methods under various imaging conditions. (a)–(d) Exhibit varying degrees of clutter noise. (e) and (f) Affected by atmospheric attenuation. (g) and (h) Experience both clutter and atmospheric interference. Cyan dashed circles in the image highlight mistakenly detected objects. Yellow dashed circles indicate undetected objects. The green dashed circle denotes the vehicle is detected with inaccurately angled bounding boxes.

the fourth row to the last, the optimal configuration of the pyramid kernel is identified, achieved through a meticulous balance of kernel size and dilation rate. The most effective APConv configuration, as detailed in the fifth row, exhibits optimal alignment with the point spread function characteristic of infrared imaging.

F. Visualization and Discussion

For a more intuitive understanding of the role played by the kaleidoscope and the SFSM in the network, we perform some visualization in this section.

1) Visualization of Detection: According to the experimental results presented in Table I, four methods with superior

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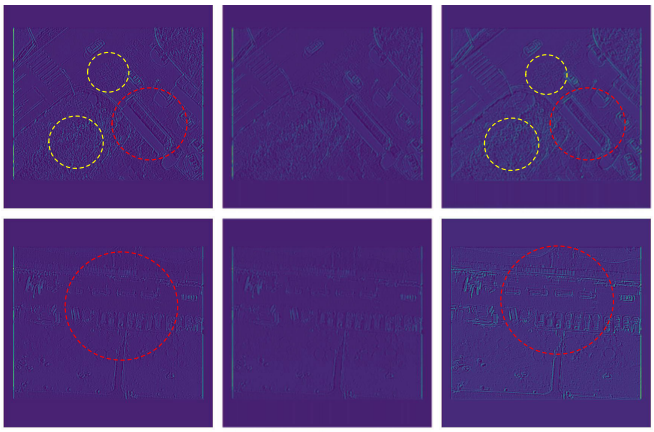


Fig. 8. Feature map visualization. (a) Input feature map, (b) feature map after guided filter processing, and (c) feature map following DGFM processing. Red dashed circles highlight vehicle edges enhanced by DGFM. Yellow dashed circles show areas where high-frequency noise has been smoothed by DGFM.

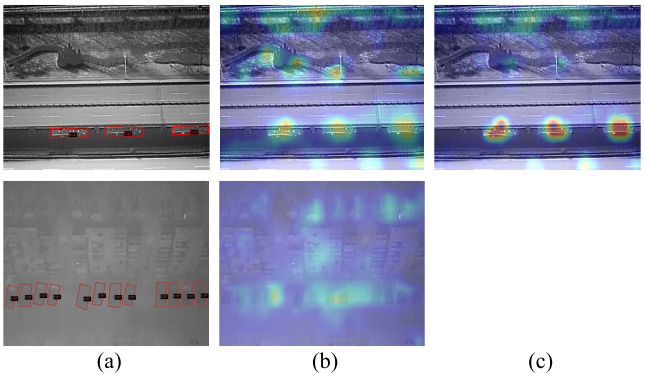


Fig. 9. Feature map visualization with and without PSRM. (a) Displays the ground truth, (b) shows the feature map without PSRM, and (c) presents the feature map with PSRM.

performance and representation are selected for output visualization: I^2MDet [9], RoI Transformer with KFIoU [33], [39], ReDet [80], and Oriented RepPoints [36]. I^2MDet achieved the best performance in infrared vehicle detection on the DroneVehicle dataset. RoI Transformer with KFIoU attained the highest score on the DOTA dataset as of July 2022. ReDet demonstrated competitive performance, boasting the second smallest number of parameters and floating-point operations. Distinct from prior anchor-based approaches, Oriented RepPoints represents objects as a set of points, which enhances flexibility in feature extraction. The output visualization of the four advanced methods and our DTNet is displayed in Fig. 7. Due to the characteristics of long-wave infrared imaging, objects often appear ambiguous in infrared images. This ambiguity causes detection methods designed for visible images to generate numerous false alarms when applied to infrared images. As depicted in Fig. 7(a) and (b), with the exception of our DTNet, the other approaches misidentify objects resembling vehicles as actual vehicles. The DGFM in our DTNet eliminates interference while preserving the edge of objects, which improves the ability of distinguishing similar items in infrared images. Atmospheric transmission degradation, which deteriorates the shape of vehicles as shown in Fig. 7(e) and (f), results in false alarms and missed detections. Although I^2MDet can recognize vehicles that are inconspicuous against the background, it struggles with delineating precise vehicle contours, as indicated by the green circle in Fig. 7(e). The PSRM in our DTNet can adaptively generate recovery functions to alleviate such degradation and some low-frequency disturbances. Consequently, with the support of DGFM and PSRM, DTNet accurately detects vehicles in infrared images, even amidst complex contextual disturbances and various atmospheric conditions.

2) Visualization of DGFM: To investigate the role of DGFM, feature maps before and after its processing are visualized. In Fig. 8, (a) represents the input feature map, (b) shows the feature map processed by guided filters, and (c) illustrates the feature map post-DGFM processing. Observations from the yellow dashed circle indicate that DGFM, similar to guided filters, diminishes high-frequency interference from road surfaces and vegetation in the feature map. As shown in the red dashed circle, DGFM not only preserves but also strengthens vehicle edges, outperforming guided filters. This visualization highlights the advanced performance of DGFM, which retains the fundamental functions of guided filters and providing additional advantages. Specifically, DGFM effectively suppresses random high-frequency interference and concurrently emphasizes vehicle edges in the feature map.

3) Visualization of PSRM: Activation regions in feature maps reflect the focal points of networks. Overlaying these maps onto original images offers insights into where networks direct attention. Feature maps from models with and without PSRM are visualized over the original image, as depicted in Fig. 9. Specifically, Fig. 9(a)–(c) corresponds to visualizations of the ground truth, the model without PSRM, and the model with PSRM, respectively. Comparing (b) and (c) reveals that the inclusion of PSRM concentrates attention more effectively toward vehicle locations, demonstrating its robustness against infrared image degradation. The first row of visualizations in Fig. 9 shows that PSRM not only enhances focus on vehicles but also suppresses background interference, illustrating its selective energy reaggregation capability.

V. Conclusion

In this article, we propose a novel framework, DTNet, specifically designed for vehicle detection in infrared aerial images. DTNet decomposes feature maps into low-frequency and high-frequency parts, processing them separately within DTBlocks. We model the infrared imaging process and develop DGFM and PSRM in DTBlocks, based on the degradation mechanism, to address challenges in infrared vehicle detection. DGFM employs learnable parameters which can effectively suppress random high-frequency interference and enhance vehicle edges in feature maps. The cascaded APConvs in PSRM are optimized via supervision to fit the point spread recovery function for adaptive energy aggregation and interference suppression. Extensive comparison experiments and ablation studies have validated the effectiveness of our designs. Additionally, we analyze the impact of various parameters on

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performance, offering insights for network design optimization. Finally, detection results and feature map visualizations of the network are presented to illustrate the contributions and enhancements of our design.

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Nan Zhang received the B.S. degree from Hainan University, Hainan, China, in 2019. He is currently pursuing the Ph.D. degree with the School of Artificial Intelligence and Automation, Huazhong University of Science and Technology (HUST), Wuhan, China.

His research interests include object detection, pattern recognition and remote sensing.



Youmeng Liu received the B.S. degree in automation from Huazhong University of Science and Technology (HUST), Wuhan, China, in 2022, where she is currently pursuing the M.S. degree with the School of Artificial Intelligence and Automation.

Her research interests include object detection and tracking.

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Hao Liu received the B.S. degree from the School of Artificial Intelligence and Automation, Huazhong University of Science and Technology (HUST), Wuhan, China, in 2018, where he is currently pursuing the M.S. degree with the School of Artificial Intelligence and Automation.

His research interests include object detection and romote sensing.



Tian Tian received the B.S. and Ph.D. degrees from Huazhong University of Science and Technology, Wuhan, China, in 2009 and 2015, respectively.

From 2012 to 2014, she was visiting Oakland University, MI, USA as a Ph.D. Student, sponsored by China Scholarship Council. She joined the School of Computer Sciences, China University of Geosciences, Wuhan, in 2015, as a Post-Doctoral Lecturer. Currently, she is an Associate Professor with School of Artificial Intelligence and Automation, Huazhong University of Science and Technology. Her major research interests include remote sensing image processing, and computer vision and applications.



Jiayi Ma (Senior Member, IEEE) received the B.S. degree in information and computing science and the Ph.D. degree in control science and engineering from Huazhong University of Science and Technology, Wuhan, China, in 2008 and 2014, respectively.

He is currently a Professor with the Electronic Information School, Wuhan University, Wuhan. He has authored or coauthored more than 200 refereed journal and conference papers, including IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE (TPAMI), IEEE TRANSACTIONS ON IMAGE PROCESSING (TIP), *International Journal of Computer Vision* (IJCV), IEEE Conference on Computer Vision and Pattern Recognition (CVPR), International Conference on Computer Vision (ICCV), and European Conference on Computer Vision (ECCV). His research interests include computer vision, machine learning, and pattern recognition.

Dr. Ma has been identified in the 2019–2021 Highly Cited Researcher Lists from the Web of Science Group. He is an Area Editor of *Information Fusion*, and an Editorial Board Member of *Neurocomputing*.



Jinwen Tian received the Ph.D. degree of pattern recognition and intelligent systems from Huazhong University of Science and Technology (HUST), Wuhan, China, in 1998.

He is currently a Professor with HUST. His research interests include remote sensing image analysis, image compression, computer vision, and fractal geometry.

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